

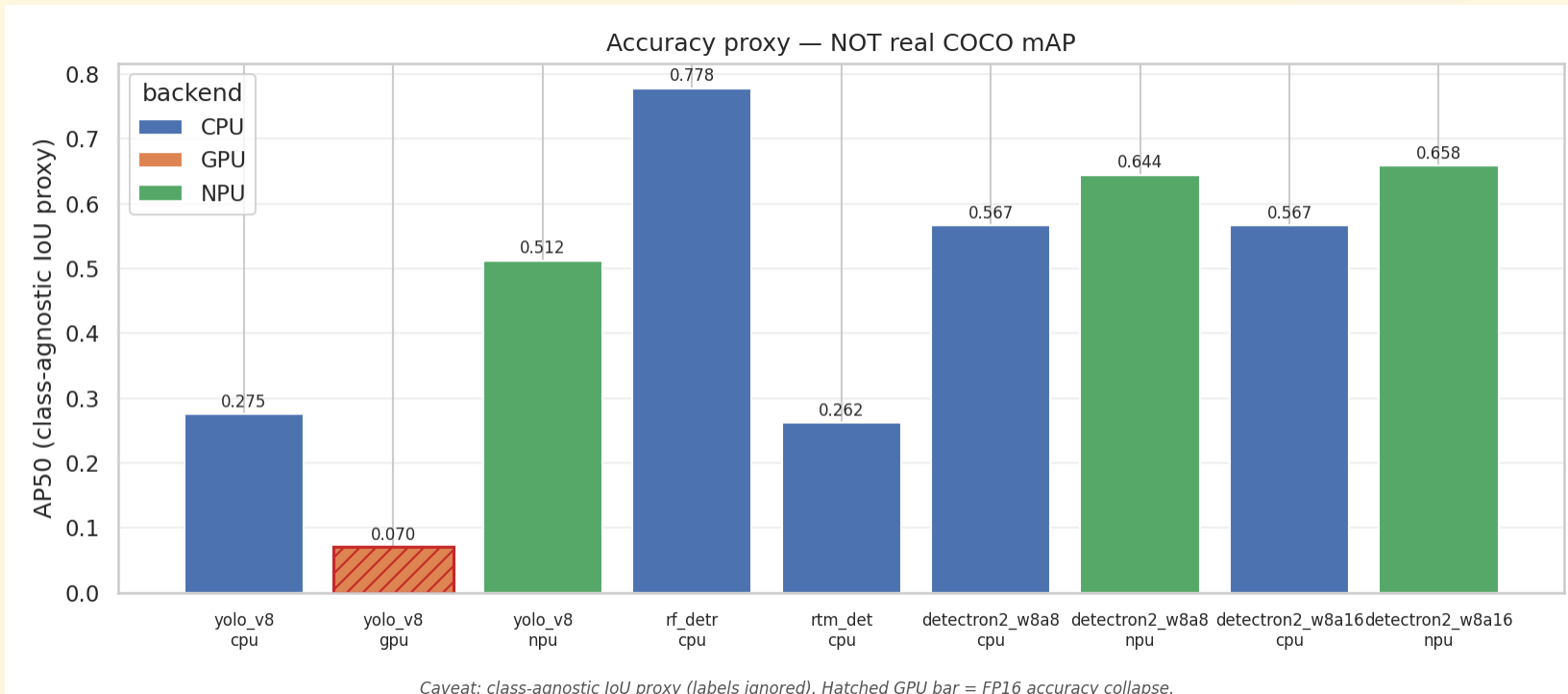
Edge Detection + LLM Reasoning on QCS6490

Update 2026-06-18

Today's update

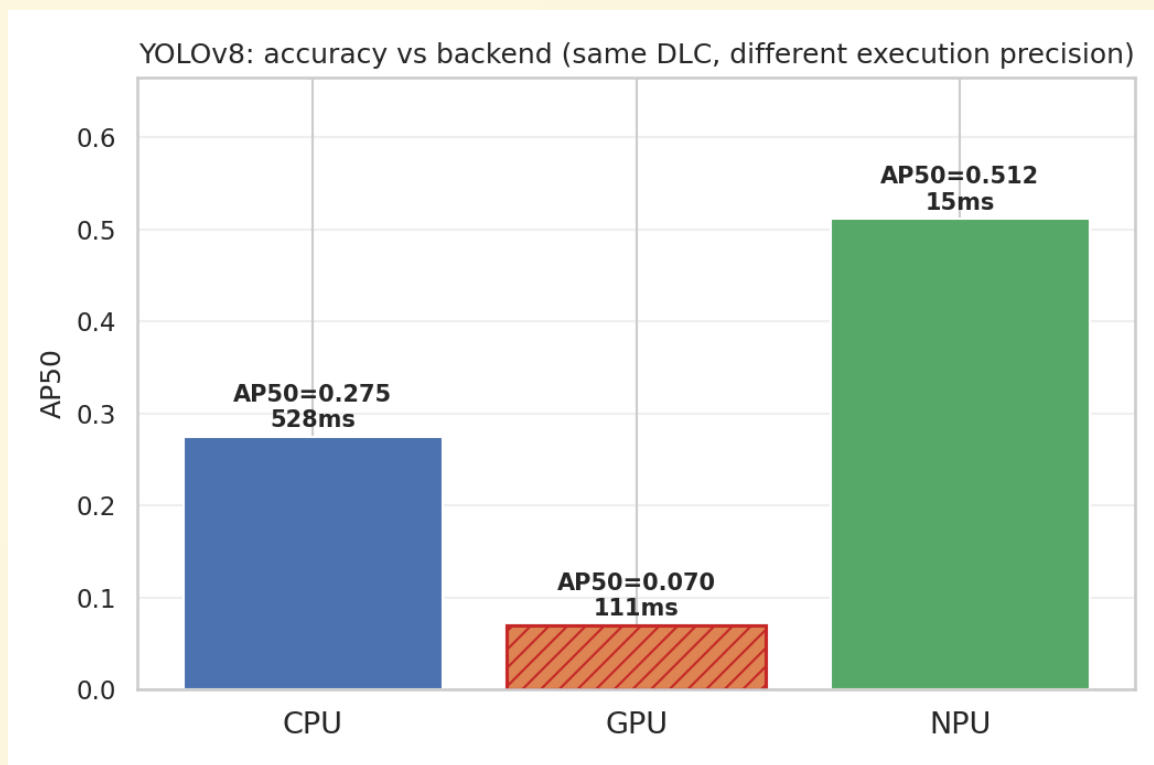
1. **Detection accuracy revisited** — full results across all models × backends
2. **YOLOv8 GPU collapse** — root cause analysis
3. **LLM text-classification benchmark** — YOLO detections → LLM → scene label

Detection accuracy (all models × backends)



RF-DETR CPU best overall (0.778). **Detectron2 NPU** best on-device balance (0.644–0.658). **YOLOv8 NPU** fastest + decent (0.512, 15 ms). **YOLOv8 GPU** collapses (0.070).

YOLOv8 GPU: what we see



GPU is fast (111 ms), but AP50 drops from 0.275 (CPU) to **0.070** – accuracy is unusable.

YOLOv8 GPU: root cause

CPU and GPU both run the **same float DLC**. NPU uses a separate context binary.

The QNN GPU backend runs in **hybrid mode**: FP32 weights are **compressed to FP16** for storage, with math accumulated in FP32.

FP16 has a 10-bit mantissa and max value ≈ 65504 . Compressing FP32 $\rightarrow$ FP16 loses precision for values not representable in that narrower format. In a deep CNN this precision error accumulates layer by layer $\rightarrow$ the output logit distribution shifts $\rightarrow$ YOLO confidence scores (passed through **sigmoid** and compared against a fixed NMS threshold) fall below the cutoff $\rightarrow$ nearly all predictions are suppressed $\rightarrow$ **AP50 collapses to 0.07**.

Evidence: collapse is fully deterministic (same images across all 3 runs); images 25 and 29 score 1.0 on GPU too — single dominant objects with logits large enough to survive the shifted distribution.

Fix (future): run GPU backend in FP32 weight mode, or re-export a DLC quantized to FP16 so compression is lossless.

LLM benchmark: setup

Pipeline: YOLO detections → structured text prompt → LLM → YES / NO / label

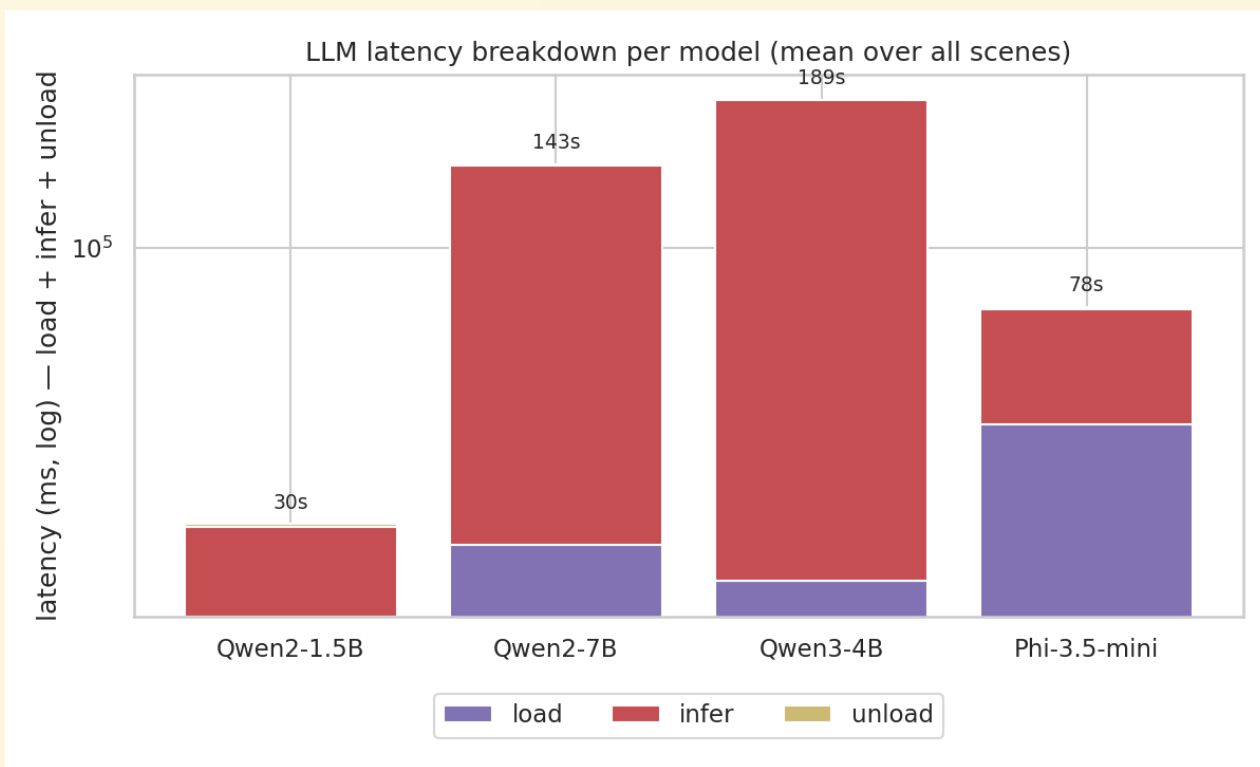
Models (CPU, via llama-server):

Model	Params
Qwen2-1.5B	1.5B
Qwen2-7B	7B
Qwen3-4B	4B
Phi-3.5-mini	3.8B

Workload: 5 apps × 2 scenes × 3 cycles (30 inferences per model)

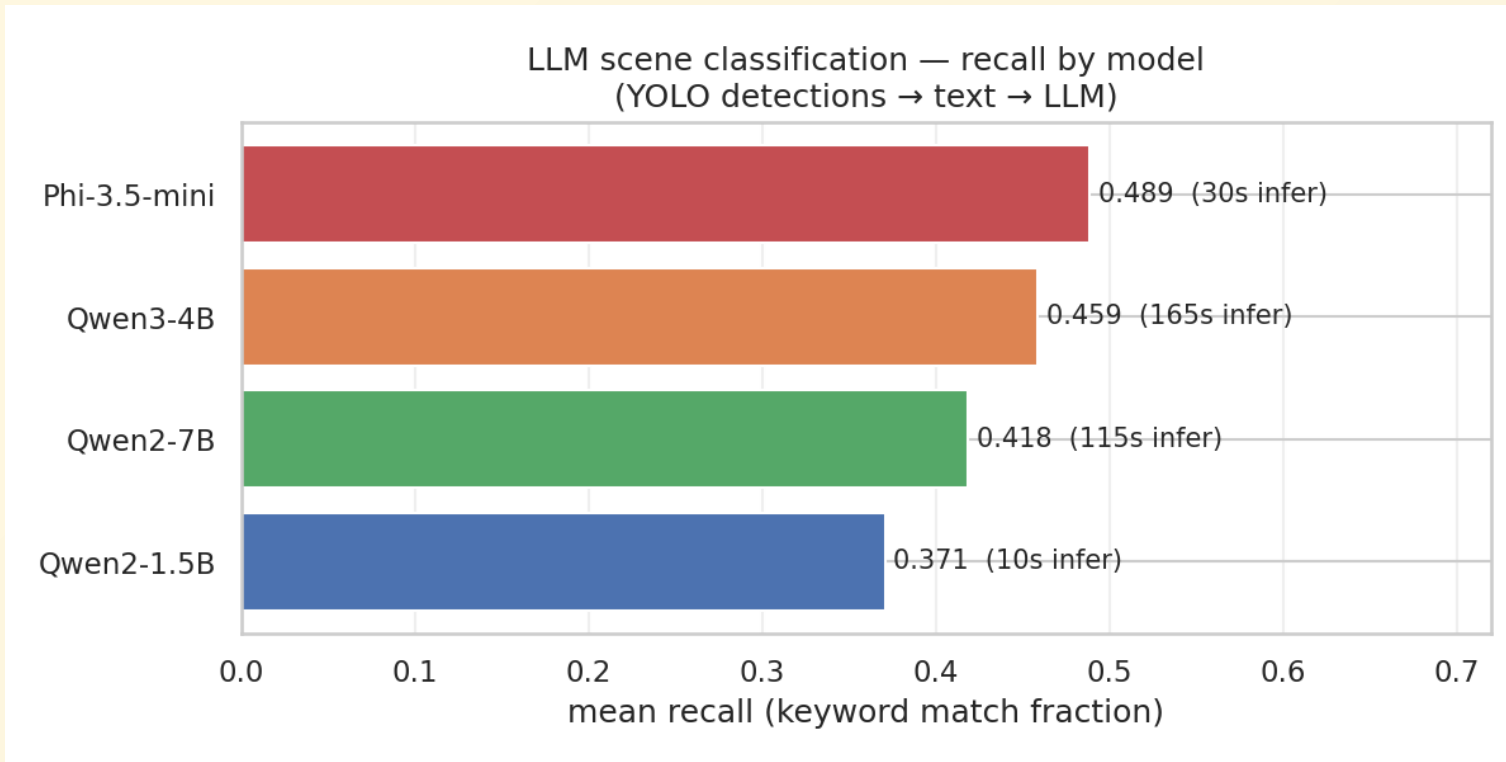
Metric: recall = fraction of expected keywords present in response (case-insensitive)

LLM benchmark: latency breakdown



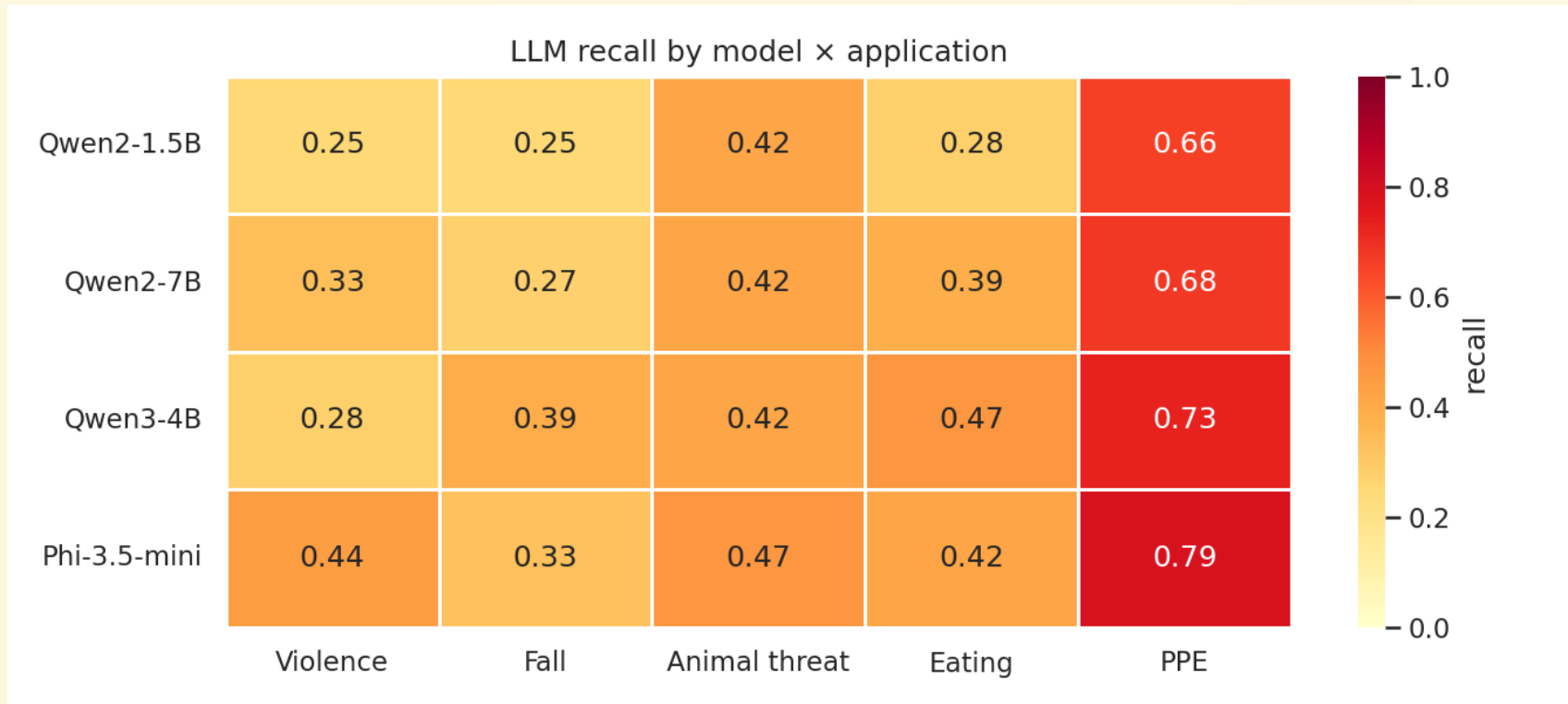
Load dominates for large models. **Qwen2-1.5B** cheapest end-to-end (≈ 30 s total). **Qwen3-4B** heaviest (> 3 min).

LLM benchmark: accuracy



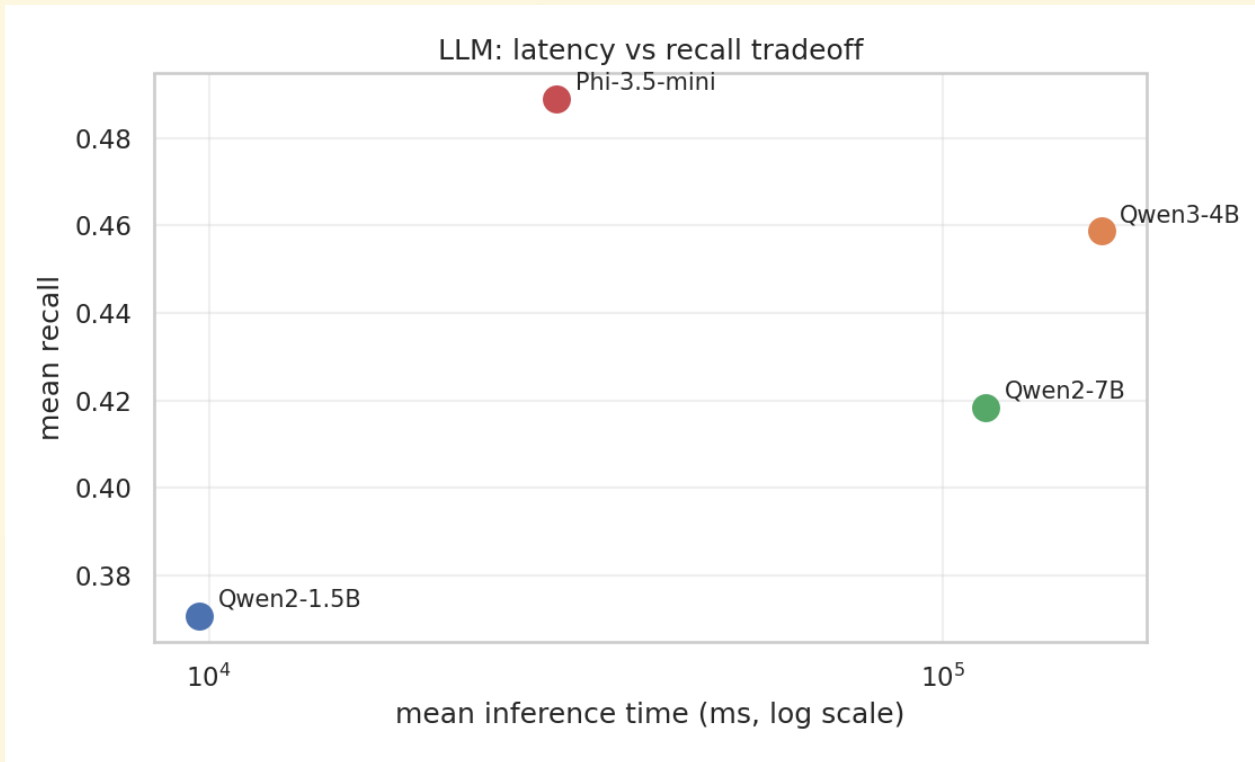
Phi-3.5-mini best recall (0.489, 30 s/infer). **Qwen2-1.5B** worst recall but **17× faster** (10 s) — edge sweet spot.

LLM benchmark: recall by app



PPE compliance easiest for all models (0.66–0.79). **Violence / fall** hardest (0.25–0.44).

LLM benchmark: latency vs recall



Qwen2-1.5B: 10× faster than Phi-3.5-mini with only -0.12 recall gap.

Qwen3-4B: 165 s/infer with no accuracy gain over Phi-3.5-mini.

Summary

Use case	Best pick	Key metric
Fastest detection (NPU)	YOLOv8 NPU	15 ms , 67 FPS
Highest detection accuracy	RF-DETR CPU	AP50 0.778
On-device detection (NPU)	Detectron2 w8a8 NPU	AP50 0.644, 377 ms
Edge LLM (speed)	Qwen2-1.5B	recall 0.37, 10 s
LLM (best accuracy)	Phi-3.5-mini	recall 0.49 , 30 s

GPU: YOLOv8 GPU fast (111 ms) but AP50=0.07 – FP16 sigmoid underflow, not usable.

Next: VLM benchmark – video frames directly into VLM (no YOLO intermediary).